CASTANEA 69(1): 38–51. MARCH 2004

Variation in Frequency of Clonal Reproduction Among Populations of *Fagus grandifolia* Ehrh. in Response to Disturbance

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ABSTRACT

Asexual reproduction has been observed to be more frequently associated with marginal habitats or with higher levels of disturbance. Here we examined frequencies of clonal reproduction across populations of *Fagus grandifolia* with varying levels of disturbance and habitat quality. We used ISSRs to identify clonal genotypes at each of five sites. Three of the sites were typical temperate habitats with relatively low densities of stems and were below 1,100 m, and two sites were above 1,370 m, possessed a high density of stems, and were characteristic of sub-alpine beech gap populations. The frequency of clonal reproduction varied substantially across sites and appears to be greater where disturbance is more severe.

INTRODUCTION

Asexual reproduction has long been recognized by ecologists and evolutionary biologists as an important factor structuring populations of many organisms (Gill et al. 1995, van Groenendael et al. 1997). This is particularly true of plants where some form of clonal reproduction can be found in almost every family. However, for both plants and animals we still know very little about how the frequency of asexual reproduction varies among populations in response to environmental variation. Furthermore, for plants, most studies have focused on clonal structure in herbaceous species, with relatively little emphasis having been placed on woody species. The majority of literature published on the mechanisms inducing clonality in trees is largely limited to silvicultural studies, which use methods of disturbance that have no real natural analog (Del Tredici 2001). In this paper we assess the association between levels of asexual reproduction in response to variation in habitat in a woody tree species.

Clonal growth has been defined in a variety of ways due to the numerous mechanisms by which it occurs. For the purposes of this work, we define clonal growth (clonal reproduction, vegetative reproduction, or vegetative growth) as a form of asexual reproduction resulting in offspring genetically identical to the parent and having the potential to become physiologically independent of the parent (Klimes et al. 1997, McLellan et al. 1997, Silvertown and Doust 1993). We refer to individual stems as ramets, while all ramets sharing the same genotype are collectively referred to as a genet (Harper 1977, Harper and White 1974). Trade-offs are often expected between vegetative growth and sexual reproduction (Harper 1977), but the extent to which this is mediated by microsite variation is not clear. Many tree species are known to produce root sprouts in response to disturbances such as herbivory, fire, flooding, and wind (Bond and Midgley 2001, Del Tredici 2001, Jones and Raynal 1988, Tourn et al. 1999). Alternatively, where habitats exist in

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environmental extremes (e.g., arctic regions, aquatic habitats, shady habitats, and nutrient-poor conditions) there is a tendency towards clonality in both herbaceous and woody plants, suggesting a possible correlation between clonality and "poor" habitat quality (Peterson and Jones 1997, van Groenendael et al. 1997). However, as pointed out by van Groenendael et al. (1997), phylogenetic constraints may be largely responsible for this apparent relationship between clonality and environment. Therefore, only within species studies of clonal population structure can clarify the role of microsite variation in the induction of clonality. The purpose of the work presented here is to examine the relationship between habitat type and frequency of clonality in a woody plant species. We use *Fagus grandifolia* Ehrh. (Fagaceae), a long-lived, geographically and ecologically widespread tree species as a test case.

In the Great Smoky Mountains (GRSM), *F. grandifolia* is found in the lower elevation temperate hardwood cove and hemlock forests, as well as in the higher elevation sub-alpine spruce-fir forests. While the cove beech forests represent a more typical and potentially more optimal habitat (loamy soils with average annual temperatures ranging from approximately 4° to 27°C for this species, the sub-alpine forests represent an environmental extreme (thin rocky soils with annual highs ~7°C cooler than coves) for the species. These forests, referred to by locals as "beech gaps," are dominated by *F. grandifolia* and occur as deciduous islands in spruce-fir forests above ~1,370 m (Russell 1953, Whittaker 1956). The beech gaps appear to be morphologically and ecologically distinct from beech stands at lower elevations, an observation that has been attributed to the extreme weather conditions that occur there (Whittaker 1956). Described as having an orchard-like appearance, the beech gaps are nearly monocultures, and are comprised of high densities of small-statured stems. This has led many observers to conclude that the beech gaps are predominately clonal in response to the marginal conditions in which they persist (Russell 1953). Additional observations that beech gaps mast (i.e., produce large seed crops) relatively infrequently when compared to low elevation stands further supports the idea that populations in these marginal habitats must be maintained by asexual reproduction (Russell 1953, Ward 1961).

Furthermore, the taxonomic identity of beech gap populations has long been the subject of debate. Camp (1940, 1950) identified three different "types" of beech based on a series of morphological characters and ecological requirements. One of the types, "white beech," Camp referred to as a coastal plain entity, and therefore it does not occur in GRSM. The other two types, "red beech" and "gray beech," both occur in, but are not limited to, GRSM. Red beech is described as being characteristic of the "acid, well drained slopes of the Appalachian uplands from Georgia northward," while the gray beech is described as "occurring at or near the margin of the spruce-fir forest ... from Nova Scotia westward to the Great Lakes region and southward along the Appalachian uplands into North Carolina and Tennessee; in the southern parts of its range it comprises the bulk of the material in the so-called 'beech orchards' ... it is restricted to alkaline soils ... one of the primary reasons why these groves are limited to the leached and less acid ridge tops.." (Camp's letter to Dr. H. J. Oosting, 20 November 1952). While there has been a more recent monograph of the genus (Shen 1992), the relationships among Camp's originally described "types" remains unclear, making it even more important to understand the ecological and evolutionary role of these populations.

A number of researchers associate root sprouting in this species with injury to the root or tree, such as that caused by foraging of mammals, freeze-thaw action, or human activity (Held 1983; Jones and Raynal 1986, 1988; Lacki and Lancia 1986). As previously mentioned, the beech gaps are typically subjected to colder winter temperatures and greater winter precipitation than the cove forests, increasing the potential for damage to roots by freeze-thaw action, and therefore (theoretically) the potential for reproduction by sprouting. However, we know of no studies in which this hypothesis has been explicitly tested. Here we use inter simple sequence repeat (ISSR) molecular markers to identify the degree of clonal structure within two types of forest, cove forests and beech gap forests, in which *F. grandifolia* is, to varying degrees, a component.

Table 1. *Fagus grandifolia* **populations sampled in Great Smoky Mountains National Park.** N **= number of individuals sampled within a plot or transect; mean DBH = mean (± the standard deviation) diameter at breast height measured in centimeters**

Site	Elevation (m)	Habitat	N	Mean DBH (cm)
Grassy Branch	466	Cove	21	2.23 ± 1.95
Greenbriar	520	Cove	20	4.32 ± 5.92
The Sinks	500	Cove	19	18.44 ± 15.27
The Chimneys	1,109	Cove	38	1.78 ± 1.77
Russell Field	1,219	Beech Gap	50	1.98 ± 1.98
Double Spring Gap	1,700	Beech Gap	50	3.76 ± 4.31
Balsam Mountain	1,625	Beech Gap	50	3.52 ± 6.37
Sweat Heifer	1,774	Beech Gap	50	3.07 ± 2.84
Cove sites (incl. Sinks)	—	—	118	5.42 ± 9.40
Cove sites (excl. Sinks)	—	—	99	2.52 ± 3.45
Beech Gap sites	—	—	200	3.08 ± 4.24
All sites	—	—	318	3.75 ± 6.32

MATERIALS AND METHODS

Study System

Fagus grandifolia occurs over a broad geographic range in the eastern United States and Mexico, where it is found in a diversity of ecological habitats. It is a shade tolerant, late successional, mesophytic tree species associated with at least 20 forest cover types in North America (Rushmore 1961). This monoecious species has male flowers occurring in catkins and female flowers occurring in clusters. Pollen is dispersed by wind. An individual tree can produce a substantial seed mast at 40 years of age, and large seed crops occur at intervals of 2 to 8 years. The primary dispersal agent for fruits is the bluejay, which can transport beechnuts across several kilometers. In addition to reproducing sexually by seed, this species can reproduce asexually by root sprouting.

Whittaker (1956) described *F. grandifolia* in the Great Smoky Mountains as being a component of the "Eastern Forest System," which includes everything except the sub-alpine spruce-fir forests and the heath balds. Within this system *F. grandifolia* is most common in mesic, non-quercine forests, which Whittaker further subdivided into cove hardwood forests, eastern hemlock forests, and gray beech (or "beech gap") forests. As *F. grandifolia* is a very minor component in true hemlock forests, we did not include them in our study. We instead focused on the frequency of clonal reproduction in *F. grandifolia* in cove forests and beech gap forests.

Field Sampling Methods

In the summer and fall of 1999, data were collected from eight populations of *F. grandifolia* in Great Smoky Mountains National Park (Table 1, Figure 1). Populations were selected to reflect representative cove forests and beech gap forests as defined by Whittaker (1956). While these two habitats cannot be explicitly defined by elevation, they appear to meet between 1,280 and 1,370 m. They are, however, quite distinct in terms of associates and overall appearances. Cove forests occur below 1,280 and 1,370 m, have a high deciduous canopy (often 30 m or more) and are dominated by various combinations of *Tsuga canadensis*, *Halesia monticola*, *Aesculus octandra*, *Tilia heterophylla*, *Acer saccharum*, *Betula allegheniensis*, *Liriodendron tulipifera*, and *F. grandifolia*. In general, the shrub layer is poorly developed, while the herbaceous layer is quite diverse. Beech gap or gray beech forests occur above 1,280 and 1,370 m, and are dominated (almost 100% in some stands) by *F. grandifolia*, but may also include *B. allegheniensis* and *A. octandra*. There is almost no shrub layer, with a limited number of species (typically sedges) in the herbaceous layer. These forests occur as "deciduous

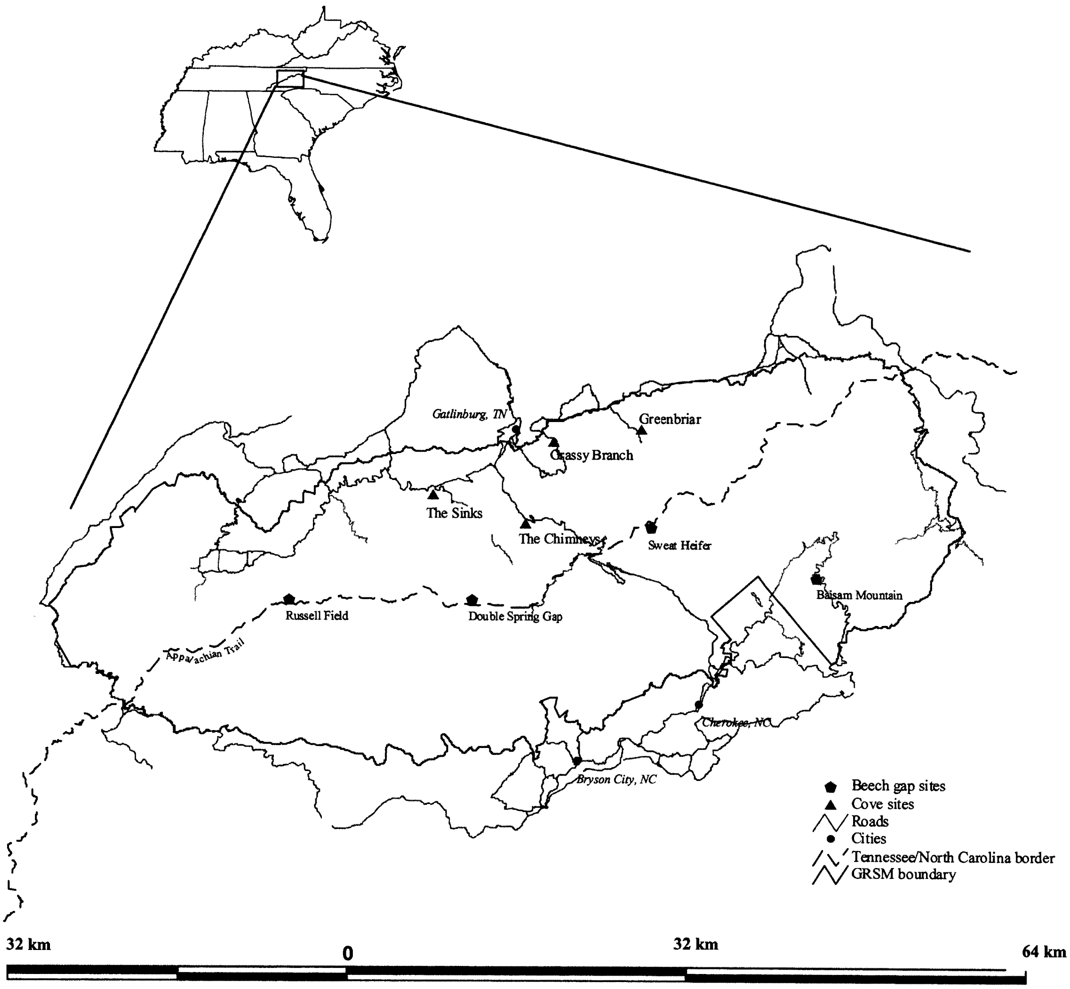

Figure 1. Sampling locations in Great Smoky Mountains National Park.

islands" within the sub-alpine spruce-fir forests of Whitaker's Boreal Forest System. Commonly referred to as "beech orchards," these forests have relatively low canopies (usually less than 12 m) and high densities of stems, giving the appearance of a monoculture.

We selected four cove sites (Grassy Branch, Greenbriar, the Sinks, and the Chimneys) and four beech gap sites (Russell Field, Double Spring Gap, Balsam Mountain, and Sweat Heifer) in which to measure the frequency of clonal reproduction. The selected cove populations are all below 1,200 m in elevation, and associated species commonly include *Tsuga canadensis*, *Aesculus octandra*, and *Acer saccharum*. In general, these sites appear upon observation to have fewer and larger stems of *F. grandifolia* than do high elevation populations. The selected beech gap populations are all above 1,200 m, and are dominated by *F. grandifolia* in what are otherwise *Picea rubens*/*Abies fraseri* forests.

Previous excavation studies indicate that clonal growth in *F. grandifolia* occurs radially from the parent tree, with most sprouts occurring within a few meters of the primary stem (Jones and Raynal 1986). Due to differences in stem densities and areal extent of the populations sampled, different sampling strategies were used in cove forest and beech gap sites. In the cove forest sites, where populations were relatively small and stems were relatively

Table 2. Clonal diversity in *Fagus grandifolia* in Great Smoky Mountains National Park. Five sites were sampled for genetic variation and clonal structure using ISSRs. Three sites were designated low elevation sites and two sites were designated high elevation sites. A total of 123 trees were assayed. Elevation indicates the approximate elevation in meters; *N* = sample size (or number of ramets; number of individual stems within a plot or transect); *G* = number of genets; *PD* = *G/N*; *MRG* = number of multiple ramet genets; largest genet = the number of stems comprising the largest genet identified; *P* = percent polymorphic loci. As *PD* approaches. 1.00, clonal reproduction is less frequent; alternatively, as *PD* approaches 0.00, clonal reproduction is more frequent

Site	Elevation (m)	Habitat	*N*	*G*	*PD*	*MRG*	Largest Genet	*P*
Grassy Branch	466	Cove	16	16	1.00	0	1	91.7
Greenbriar	520	Cove	23	20	0.87	2	3	87.5
Chimneys	1,109	Cove	19	8	0.42	2	10	45.8
Balsam Mountain	1,625	Beech Gap	30	22	0.73	3	8	83.3
Double Spring Gap	1,700	Beech Gap	35	34	0.97	1	3	91.7

sparse, a 10 m × 20 m plot was established for sampling, resulting in a variable number of samples in each population (Table 1). At Grassy Branch, Greenbriar, and the Sinks, all *F. grandifolia* stems within the plot were sampled. At the Chimneys, where the population was small but stem density was high, 12, 1 m^2 subplots were randomly selected within the plot using a grid system and a random number generator. In each subplot, all *F. grandifolia* stems were sampled. In the beech gap sites, where populations were extensive and stem densities were high, a 90 m transect was established, along which samples were collected at 10 m intervals. At each sampling interval the five closest stems were sampled and all sampled stems were mapped relative to plot boundaries, resulting in 50 sampled stems in each population (Table 1). We chose to use a transect method in the beech gap sites because we wanted to identify as many unique genets as possible. Assuming that clonal structure in *F. grandifolia* is radial as reported by Jones and Raynal (1986), and observing that stem density is much greater in the beech gaps than in the cove forests, a transect would be more likely to cross more genets than would a plot or grid based approach. Alternatively, in the cove populations, stem density and total number of stems were limited enough that a transect approach was not feasible.

In all populations, sampled stems were mapped relative to plot (or transect) boundaries, and were measured for diameter at breast height (DBH). For stems less than 1.5 m in height, diameter was measured at the base of the stem using a 152 mm dial caliper. No size class was excluded from sampling. Leaf material was collected from each stem sampled, placed in separate 1.5-mL microcentrifuge tubes, and stored on ice while in the field. Upon returning to the lab, all samples were snap frozen in liquid nitrogen and stored at −70°C.

ISSR Protocol

A subset of populations was sampled for genetic structure using inter-simple sequence repeat (ISSR) markers (for a review see Wolfe and Liston 1998) (Table 2). We included three cove populations (Grassy Branch, Greenbriar, and Chimneys) and two beech gap populations (Double Spring Gap and Balsam Mountain). Due to loss of frozen leaf material because of freezer failure in the spring of 2000 and some generally problematic samples, the actual sample size for genetic analysis was not equal to the sample size used for diameter analysis (Tables 1 and 2).

Total genomic DNA was extracted from each leaf sample using a protocol adapted from Edwards et al. (1991) as modified by Martin and Cruzan (1999). Each sample was cleaned by binding the DNA to DEAE-cellulose as described by Marechal-Drouard and Guillemaut (1995) and then stored at −20°C.

A number of ISSR primers from the University of British Columbia primer set #9 were initially screened for variability and reproducibility. Primer 890 (VHV GTG TGT GTG TGT GT) was selected for further analysis given the large number of loci generated, consistency of band intensity, and reproducibility across populations. A subset of samples was replicated to insure repeatability of banding patterns. Single-primer amplifications were carried out in 15 μl volumes as follows: 2.5 mM $MgCl_2$, 200 μM dNTPs, 1 unit of *Taq* polymerase, 0.10 μM primer, and 0.5 μl DNA. The thermal cycler profile was adopted from Huang and Sun (2000): 1 cycle of 94°C for 5 minutes, followed by 45 cycles of 94°C for 5 seconds, 50°C for 45 seconds, 72°C for 1.5 minutes; and a final 7 minute extension at 72°C. PCR products were separated on 2% agarose gels in 1X TBE buffer until the bromphenol blue marker migrated 10 cm from the origin. Gels were stained with ethidium bromide and were documented digitally using Kodak 1D Biomax software. Bands with the same molecular weight were treated as identical loci. A data matrix was compiled in which loci were scored as 1 or 0 for band presence or absence, respectively. The size of scored loci ranged from approximately 300 bp to 1,500 bp. Bands greater than 1,500 bp were not scored due to their ambiguity and inconsistency. Ramets were treated as members of the same genet if they shared identical banding patterns. A difference of one or more bands was used to distinguish between genets.

Sampling Efficiency

Two methods were used to determine if sampling more loci were necessary to determine whether the majority of genetic variation had been detected. The first method, introduced by Aspinwall and Christian (1992) calculates the average probability that two spatially separated ramets belong to the same genet:

$$\overline{P} = 1 - \frac{\left[\sum_{Q=1}^{N} \prod_{D=1}^{M} \left[\frac{X_{DQ}}{P_D}\right]\right]}{N}$$

where Q is the particular individual in a sample, N is the total number of individuals in a sample, M is the number of polymorphic loci in the sample, D is an individual polymorphic locus, X_{DQ} is the number of individuals in the sample of the same Dth multi-locus genotype as the Qth individual, and P_D is the total number of individuals examined for each primer. There is an asymptotic relationship between this probability and the number of polymorphic loci in a population such that as the probability approaches 1, additional assays will result in diminishing returns.

In the second method, varying numbers of polymorphic loci (1 through the maximum number found) were randomly resampled to infer the effects of increasing the number of markers on the number of genets detected. The number of genets was calculated for one thousand replicates of each data set size and these data were used to examine the relationship between number of loci in the data set and the number of genets detected. We assumed that an asymptotic curve (i.e., the number of new genets detected approaches zero as the number of loci in the data set is reached) would indicate that adding additional loci would not result in an appreciable change in the total number of genets detected.

Estimates of Clonal Diversity

Clonal diversity was estimated for each site using the 'proportion distinguishable' ($PD = G/N$, where G is the number of genets and N is the total number of ramets), an overall descriptor of diversity proposed by Ellstrand and Roose (1987). The relative importance of sexual and asexual reproduction was estimated by calculating mean clonal identity at 5 m intervals for each site (Harada and Iwasa 1996, Harada et al. 1997, Schlapfer and Fischer 1998) (Figure 2). This was accomplished by comparing all possible pairs of ramets within a site. Distances between pairs were determined from site map data. If the pair shared the same multi-locus genotype, a clonal identity of 1 was assigned. If the pair did not share the same multi-locus genotype, a clonal identity of 0 was assigned. The mean clonal identity was then calculated for each of six distance classes at intervals of 5 m each (i.e., 5, 10, 15, 20, 25, and 30 m). For example, the first distance class represents the mean probability of clonality among all possible pairs of ramets separated by less than 5 m within a site. To compare the clonal genetic structure

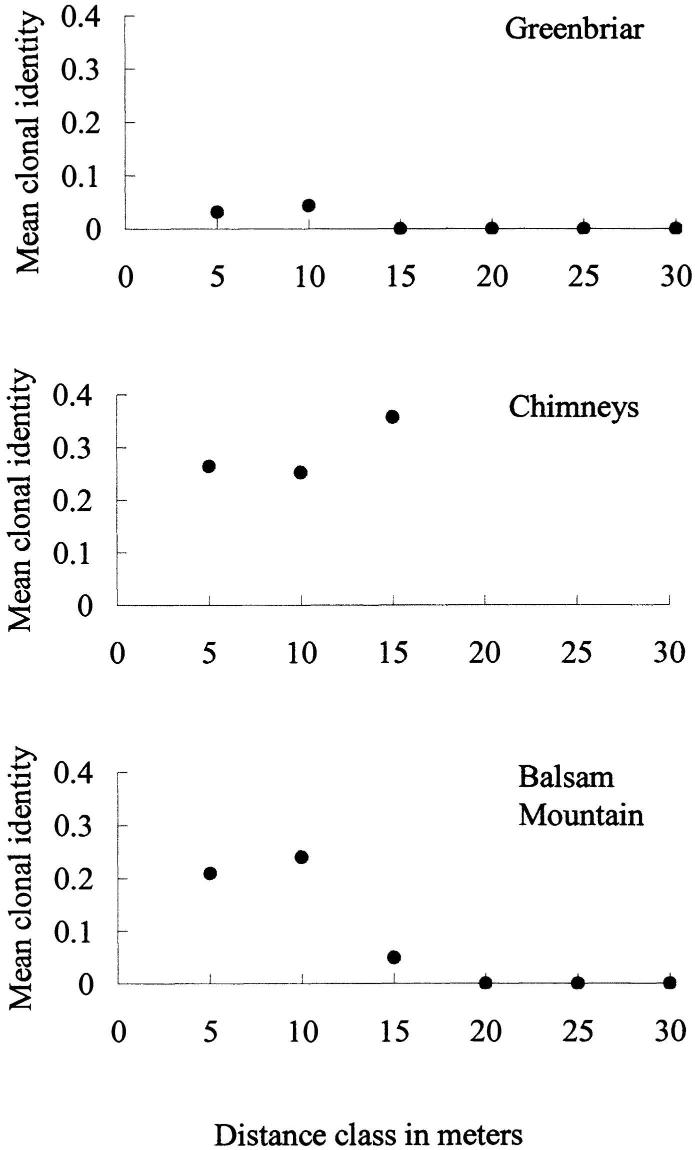

Figure 2. Mean clonal identity by site for *Fagus grandifolia* in Great Smoky Mountains National Park. Mean clonal identity was calculated for each of six distances classes at intervals of 5 m each. As mean clonal identity approaches 0.00, sexual reproduction is assumed to be dominant over clonal reproduction. Alternatively, as mean clonal identity approaches 1.00, clonal reproduction is assumed to be the dominant mode of reproduction. In theory, mean clonal identity decreases with increasing distance between ramets.

among populations, the relationship between mean clonal identity and distance was then plotted for each population. In addition, to test the potential effects of somatic mutation on clone identification a neighbor-joining phylogram based on the distance coefficient of Nei and Li was constructed using PAUP* 4.0b10 (Swofford 1999) (Figure 3).

Estimates of Genetic Diversity

Gene diversity within and among populations was estimated using the program AMOVA 1.55 (Excoffier 1995). This program calculates components of molecular variance at three

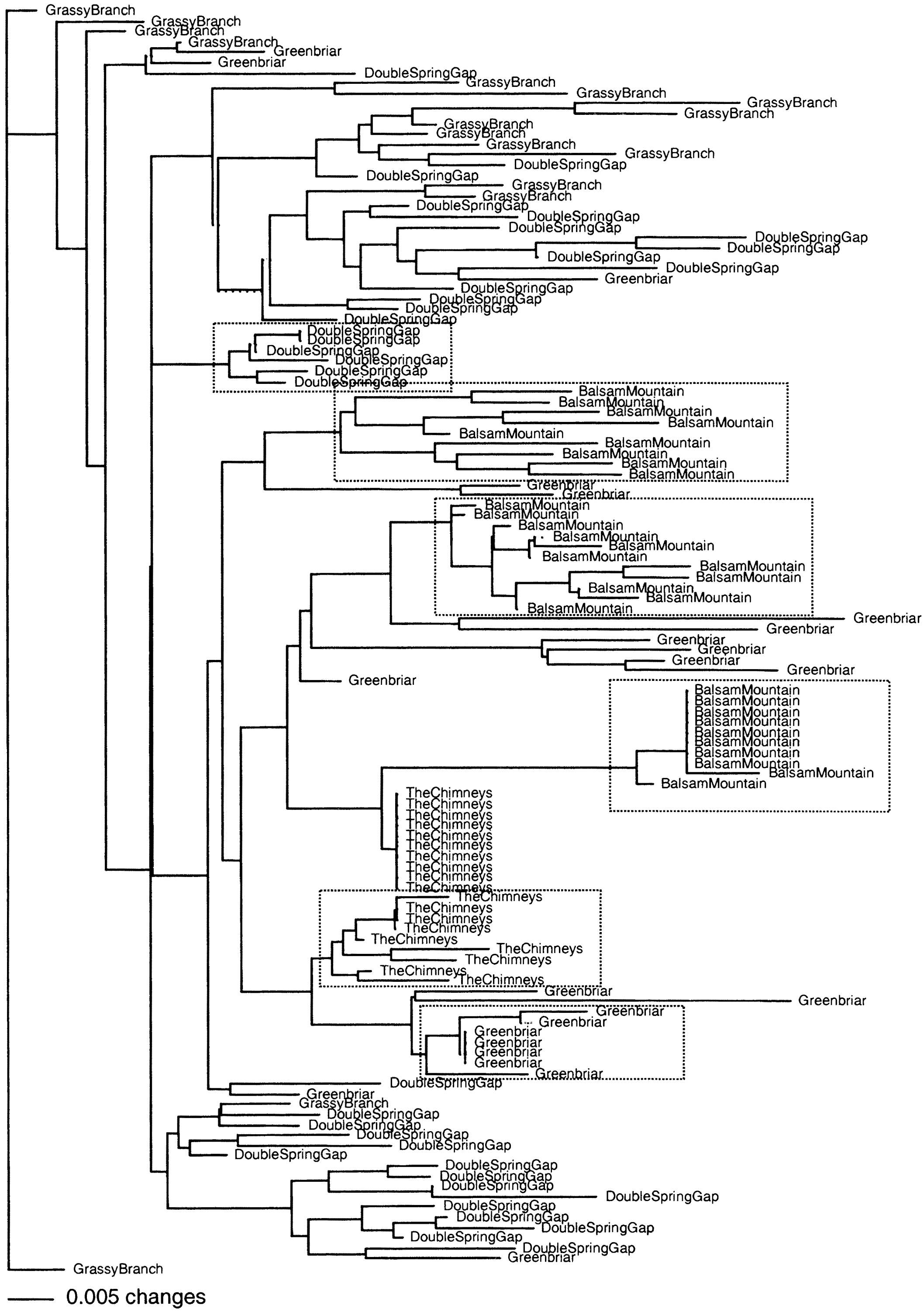

Figure 3. Neighbor-joining phylogram of *Fagus grandifolia* based on ISSR data. Boxes represent potential multi-ramet genets obscured in ISSR data by somatic mutations.

hierarchical levels: among groups (i.e., habitat), within groups (i.e., among populations within groups), and within populations (i.e., individuals).

RESULTS

Data Efficiency

Analyses of ISSR marker efficiency indicate the genet detection was close to saturation (Table 2). Our estimates of P were consistently high, 0.99 across all sites, and was never less than 0.98 within a site (Table 2), indicating that there was a low chance of assigning ramets to the wrong genet. Resampling of loci indicated that the relationship between the number of loci used and the number of genets detected was asymptotic, so additional loci would be unlikely to identify more genets. Both methods indicate that a sufficient number of polymorphic loci was obtained to efficiently assess clonal genetic diversity within and among sites with a high level of confidence.

Clonal and Genetic Diversity

A total of 123 ramets was assayed for genetic variation across five sites (Table 2). The number of ramets per site ranged from a minimum of 16 to a maximum of 35. The ISSR primer used revealed 24 loci, 23 of which were polymorphic. One hundred unique genets were identified across all sites. The number of genets per site ranged from a minimum of 8 to a maximum of 34. Genetic diversity was significant (p-value < 0.001) within and among populations, explaining 75.66% and 24.34% of the total variation, respectively (Table 3). There was no significant difference between cove and beech gap populations (p-value $= 0.8860$), explaining less than 5% of the total genetic variation observed (Table 3).

There were substantial levels of clonal diversity both within and among sites of *F. grandifolia* (Table 2). The mean genetic polymorphism within sites was 80% of loci sampled, ranging from a minimum of 45.83% to a maximum of 91.67%. The total genetic polymorphism across all sites was 95.8% of the loci sampled. The proportion distinguishable (*PD*) ranged from a minimum of 0.42 in the Chimneys population to a maximum of 1.00 in the Grassy Branch population. All genets were local, meaning that no individual genet occurred in more than one population. Most genets consisted of only a single ramet.

Mean clonal identity was estimated for three (the Chimneys, Greenbriar, and Balsam Mountain) of the five sites (Figure 2). The two remaining sites (Grassy Branch and Double Spring Gap) were predominately sexual ($PD = 1.0$ and 0.97, respectively), making the calculation of clonal identity unnecessary. The greatest distance between two ramets of the same genet was 16.25 m and occurred at Balsam Mountain. Mean clonal identity was greatest across sites within the 15 m distance class for the Chimneys, with a value of 0.357. At this particular site, mean clonal identity continued to increase with increasing distance between ramets (Figure 2). Because we would expect mean clonal identity to decrease as distance increases, this indicates that the identification of additional clone mates beyond the sampling area is likely, suggesting the possibility of a much more extensive genet at this site. At both Grassy Branch and Balsam Mountain, mean clonal identity is greatest within the 10 m distance class, with values of 0.044 and 0.240, respectively. Beyond the 10 m distance class, mean clonal identity was 0.00 at both sites, indicating that additional ramets beyond the sampling area are not likely.

The neighbor-joining phylogram reflects no clear pattern of structuring among populations (Figure 3). There are, however, groups of individuals from each population (except Grassy Branch) that form small clades of closely related individuals that were not identified by ISSRs as members of the same genet.

The Effect of Habitat on Diameter

Because of the limited sample sizes, we did not perform any formal statistical analyses on our diameter data. We compared the mean diameter values among habitats to identify trends in the data (Table 1). When comparing the mean diameter of all cove sites (5.42 ± 9.4) to that of all

Table 3. Analysis of molecular variance (AMOVA) for 123 individuals of *Fagus grandifolia* based on ISSR markers. Statistics include sums of squared deviations (SSD), mean squared deviations (MSD), an estimate of variance, the percent of the total variance explained by each component, and the p-value associated with each component

Source	d.f.	SSD	MSD	Variance	% Total variance	p-value
Between habitats (Coves vs. Beech gaps)	1	16.4808	16.48	0.1954	4.16	0.8860
Among sites	4	128.4188	32.105	1.1799	24.34	<0.001*
Within sites	118	432.6869	3.667	3.666	75.66	<0.001*

beech gap sites (3.08 ± 4.24), there is a trend toward smaller stems in the beech gap sites. However, we noted that the Sinks population had much larger stem sizes on average than did the other cove sites. After removing the Sinks population from the data, the overall mean of the cove sites decreased significantly (2.52 ± 3.45), such that the observed trend was reversed (smaller stems in the cove sites). Therefore, we concluded that these data were not robust enough to test Whittaker's (1956) early observations that beech gaps exhibit smaller stem sizes than do cove forests.

DISCUSSION

The objective of this work was to test the long held assumption that beech gaps are maintained largely by clonal reproduction. To do this, we compared the genetic structure of beech gap populations to those of cove populations of *Fagus grandifolia* in GRSM. Our findings suggest that habitat type is not predictive of the frequency of clonality within a site. There was a great deal of variation in frequency of clonality among sites, irrespective of habitat type (Table 2). Sexual reproduction appears to be common in the beech gaps, which is inconsistent with previous hypotheses.

Variability in Frequency of Clonal Reproduction

Our results indicate that clonal reproduction is not dependent on habitat type. Alternatively, the high level of clonality observed in only one of the study populations (the Chimneys) may in fact be attributable to an introduced pathogen. Of all sites sampled, the Chimneys had the lowest *PD* value (0.42), indicating the highest proportion of clonality within a population. This is the only site in this study that has been severely affected by beech bark disease. This disease, a combination of scale insect and ascomycete fungus, was introduced into Nova Scotia in the late 1800's and has since spread southward throughout most of New England (Ehrlich 1934, Houston et al. 1979). GRSM represents the southern-most occurrence of the disease, which was first recorded there in 1993 after the observation of extensive canopy decline in stands of *F. grandifolia* (Blozan 1995). While all of the beech gap sites appear to exhibit early stages of infection, the Chimneys site is the only site from the present study in which damage as a result of the disease has been extensive (pers. obs.). In previous long-term monitoring studies of beech bark disease in GRSM, mortality of adult stems at the Chimneys site increased 30% over a 4 year period, with the greatest increase (from 15% to 30%) occurring in the last year of the study (Wiggins 1997). Other studies indicate that the larger, older stems are the first to die in response to beech bark disease, and the root systems of these stems respond by prolific sprouting (Ostrofsky and Houston 1988). DiGregorio et al. (1999) recently noted the importance of beech bark disease as a diffuse disturbance, citing the role of dead and dying trees as abundant gapmakers in infected stands. While their interests focused on the response of *Acer saccharum* to diseased gapmakers, their conclusions regarding increased growth in the sapling layer should apply to other shade tolerant species as well, further supporting the idea that clonally reproduced sprouts may rapidly repopulate the sapling layer following beech bark disease-induced mortality.

While we cannot verify from which stem a given genet originated, we did observe that the majority of the largest stems at the Chimneys were already dead in response to this disease.

This observation, in conjunction with the observation that a large proportion of the ramets sampled at this site were less than 1.5 m tall, suggests that the extensive clonality observed at this site is likely a recent response to this introduced disease. The Chimneys may therefore represent a site in which a non-native pathogen has resulted in a shift from sexual to asexual reproduction in response to damage inflicted on the canopy trees.

The Balsam Mountain site, while limited in the overall frequency of clonality ($PD = 0.73$), had the second largest genet (8 ramets) recorded in this study. This genet was composed of two ramets greater than 20 cm DBH (both of which were highly infected with beech scale), and six relatively small ramets (all less than 1.5 cm DBH). Jones and Raynal (1986) found no relationship between root suckering and beech bark disease infection as long as the parent tree was still alive, therefore, it is not likely that the disease is the cause of the clonality observed at this site. One potential explanation is the tendency for wild pigs (*Sus scrofa*; introduced into the park in the 1940s) to selectively forage in the beech gaps. Previous studies by Lacki and Lancia (1986) identified increased root sprouting in beech gaps frequented by pigs, which they attribute to enhanced litter decomposition and soil nutrient mobilization caused by pig rooting. While we did not take note of rooting in the Balsam Mountain site, it is not unusual to see such disturbance when visiting beech gap populations, making it a viable explanation for the limited amount of clonality observed in that particular site.

Finally, we must address the fact that Greenbriar, a site that has not yet been infected by disease, exhibits some limited amount of clonality ($PD = 0.87$). This site has two multi-ramet genets, one with 2 ramets and one with 3 ramets. We suggest that this level of clonality can be explained by stochastic disturbance. Greenbriar is a somewhat rocky site adjacent to the Little Pigeon River, which is subject to springtime floods. Some of the stems sampled at this site were hanging over the river, with roots exposed along the bank, making them susceptible to abrasion by rocks and debris. Therefore, we suggest that root abrasion is a viable hypothesis to explain the limited amount of clonality observed at this site.

Somatic Mutation

One concern of clonal ecologists using molecular markers is that genet size may be underestimated as a result of somatic mutations. With fragment-based markers such as ISSRs, using a single band difference criterion is the least subjective way to differentiate between genets. However, this approach may incorrectly identify a ramet that is actually a member of a multi-ramet genet as a unique genet due to the presence of somatic mutations. Here we approached this problem by constructing a neighbor-joining phylogram to investigate the relationships among individuals (Figure 3). While individuals do not generally group by population, there are several small clusters across the phylogram that are comprised entirely of individuals from one population. In most cases these clades, which are all constructed of relatively short branches, include multi-ramet genets recognized by ISSRs. Within each of these clusters, the greatest geographic distance between individuals is 10 m, with most of the stems being less than 2 m apart. In the event that somatic mutations have occurred among ramets within a clone, these ramets would likely appear as sister to each other within the same cluster, just as we observe here. We raise the possibility that clonality is more extensive in these populations than has been detected by the molecular markers used. A thorough analysis of frequency of somatic mutation in *F. grandifolia* is necessary to verify this conclusion. Interestingly enough, this observation does not nullify our conclusions regarding disturbance induced clonality. The phylogram presented here indicates that all of the individuals sampled in the Chimneys population, which was the most affected by disease of all sampled populations, could potentially represent only two genets (Figure 3). In addition, we see the potential for much higher levels of clonality in the Balsam Mountain population, which we hypothesized has been disturbed by wild pigs. There also appears to be the potential for more clonality at Greenbriar, the low elevation site that is susceptible to disturbance by flooding. Therefore, disturbance (of various types) continues to be a valid explanation for increased clonality in this long-lived woody species.

Limited Differentiation Across Elevations and Among Sites

Genetic structure was greater within than among sites but there was no evidence for strong differentiation between the coves and beech gaps studied here. This observation suggests that the apparent geographic isolation of beech gap populations from cove populations of *F. grandifolia* is not sufficient to prevent gene flow among sites. Again, referring to the phylogram, it is apparent that all of the individuals sampled are relatively closely related, with very little change in similarity recognized across the full length of the tree (Figure 3). This is consistent with the fact that *F. grandifolia* is a wind pollinated species, which makes effective population sizes much larger. It would be useful in future studies to test the relative contributions of seed and pollen in these populations using both nuclear and organellar markers. Such a study would verify whether or not pollen flow is the factor limiting genetic structure across this landscape, as well as which populations (e.g., cove or beech gap) appear to contribute more seeds and pollen to the gene pool.

Conclusions

We found disturbance, both from an introduced pathogen and natural flooding, to be the most likely factor driving frequency of clonality within a site, as opposed to the ecological differences of coves and beech gaps, as were previously assumed. While several recent studies have emphasized the importance of environmental stability as a factor leading to the production of larger and fewer genets in natural plant populations (Xie et al. 2001), disturbance is less commonly cited as a contributing factor (but see Gom and Rood 1999). The work presented here provides new insights into the conditions that may favor asexual reproduction, as well as emphasizing the need to recognize somatic mutation within genets. In addition, this work highlights the importance of incorporating both environmental variation and different levels of disturbance into studies of the distribution of clonal reproduction in woody plant species.

ACKNOWLEDGMENTS

The authors thank C. Morris, G. Wiggins, G. Taylor, M. Fischer, S. Datwyler, and M. Gitzendanner for analytical and logistical assistance during the course of this research. This manuscript is part of a Master's thesis completed at the University of Tennessee, and was funded by a Campbell Fellowship from the Great Smoky Mountains Conservation Association.

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Received July 30, 2002; Accepted March 3, 2003.